

JAYACHANDRA GALDA

Buffalo, NY | jayachandragalda@gmail.com | (716) 295-3762

GitHub: github.com/Jayachandragg | LinkedIn: [linkedin.com/in/jaya-chandra-galda-077aa624b](https://www.linkedin.com/in/jaya-chandra-galda-077aa624b) | Portfolio: <https://jayachandragg.github.io/tensor-essence/>

PROFESSIONAL SUMMARY

AI/ML Engineer with an M.S. in Artificial Intelligence from the University at Buffalo and hands-on experience building production ML systems across healthcare, research, and industry settings. Skilled in RAG-based LLM architectures, computer vision pipelines, and cloud-native ETL workflows using Python, PyTorch, FastAPI, and AWS, with a track record of measurable impact including cutting verification latency by 60% and achieving 96%+ detection accuracy in deployed systems. AWS Certified Data Engineer – Associate, focused on shipping scalable, real-time AI systems from research to production.

EDUCATION

University at Buffalo, State University of New York

M.S. in Engineering Science (Artificial Intelligence)

Jan 2025 – May 2026

- Relevant Coursework: Machine Learning, Reinforcement Learning, Computer Vision, Pattern Recognition, Database Management Systems, Analysis of Algorithms

Mahindra University, India

B.Tech in Computer Science and Engineering

Jan 2020 – Jan 2024

SKILLS

- Programming Languages: Python, SQL, C/C++, JavaScript
- Machine Learning & Vision: PyTorch, TensorFlow, Scikit-learn, YOLOv8, OpenCV, SqueezeNet, MLflow, Sentence-Transformers, pypmi
- LLM & GenAI: RAG Architectures, Llama-3.3-70B, Qwen2.5-0.5B-Instruct, Vision-Language Models (VLMs), Hugging Face Transformers, Groq API, FAISS, Prompt Engineering
- Data & Cloud Engineering: PostgreSQL (3NF), MongoDB, FastAPI, Uvicorn, Apache Spark, AWS (EC2, S3, Lambda, ECS), Docker, REST APIs, ETL Pipelines, GitHub Actions
- Deployment & BI: Hugging Face Spaces, Streamlit, Gradio, Tableau, Power BI

WORK EXPERIENCE

QAS.AI | AI & Automation Engineer | Buffalo, NY | [GitHub](#)

Jan 2026 – Jun 2026

- Engineered an automated 7-step clinical decision engine (Python, Pandas, NumPy) converting AHA/ACC medical guidelines into a sensor-driven rule engine for medication titration.
- Architected a multi-stage ETL pipeline in SQL to transform ~500k raw hospital records from the MIMIC-IV dataset down to 1.6M rows (6,319 patient records) of structured, queryable sensor and lab data.
- Deployed a RAG-based conversational endpoint utilizing FastAPI, Groq Llama-3.3-70B, and FAISS vector indices, achieving real-time document retrieval and knowledge base Q&A.

University at Buffalo | Graduate Research Assistant | Buffalo, NY | [GitHub](#)

Aug 2025 – Jan 2026

- Engineered StrategyGen, an end-to-end multimodal video segmentation pipeline converting raw ELAN (.eaf) behavioral annotations into clean, event-level JSON data.
- Implemented a hybrid fusion pipeline combining deterministic rulebased merging (pypmi) with a CPU-optimized SLM (Qwen2.5-0.5B-Instruct) for semantic boundary and transcript refinement.
- Designed gap constraint guardrails and fallback logic to process noisy annotations, saving ~20 operational hours/week and cutting verification latency by 60%.
- Deployed an interactive demo on Hugging Face Spaces (Gradio + PyTorch) enabling CPU-friendly inference for multimodal ML preprocessing.

Digitap.AI | Machine Learning Engineer | Bengaluru, India

Jan 2024 – Aug 2024

- Deployed an end-to-end real-time face, eye, and mask detection pipeline using YOLOv8 + OpenCV, achieving a 96.3% detection accuracy rate across high-throughput data streams.
- Reduced inference latency by 23% through model optimization and batching; engineered a lightweight SqueezeNet-based CNN for grayscale face detection (98% accuracy) on low-compute edge setups.
- Developed a 3D facial alignment algorithm across X, Y, and Z axes, boosting face verification success rates by 21%, and built a YOLOv8 blur-detection classifier (95.4% accuracy).
- Implemented an aspect-ratio visibility filter to drop occluded faces, improving training dataset quality by 18% and decreasing downstream alert fatigue.

IIT Roorkee | Data Analytics Research Intern | Roorkee, India

Sep 2023 – Dec 2023

- Benchmarked multiple deep learning and neural architectures across a 53-category operational dataset.
- Constructed a hybrid ensemble model that achieved an ~8% improvement in baseline evaluation metrics (F1-score).

PROJECTS

Unsupervised Deepfake Detection via β -VAE & Latent Classification

- Trained a β -Variational Autoencoder ($\beta=4$, 128-dim latent space) on 6,800+ real and synthetic (GAN/Diffusion) images to extract disentangled feature representations in a self-supervised manner.
- Engineered a downstream Logistic Regression classifier utilizing frozen latent vectors, attaining a 98% accuracy rate and 0.997 ROC-AUC while decoupling representation learning from task-specific classification.

E-Commerce Data Analytics Platform

- Modeled a 3NF PostgreSQL transactional database schema to convert raw log records into operational KPIs and business insights.

CERTIFICATIONS

- AWS Certified Data Engineer – Associate (AWS)